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Five lessons for avoiding failure when scaling in conservation

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Many attempts to scale conservation actions have failed to deliver their intended benefits, caused unintended harm or later been abandoned, hampering efforts to bend the curve on biodiversity loss. Here we encourage those calling for scaling to pause and reflect on past scaling efforts, which offer valuable lessons: the total impact of an action depends on both its effectiveness and scalability; effectiveness can change depending on scale for multiple reasons; feedback processes can change socio-ecological conditions influencing future adoption; and the drive to scale can incentivize bad practices that undermine long-term outcomes. Cutting across these themes is the recognition that monitoring scaling can enhance evidence-informed adaptive management, reporting and research. We draw on evidence and concepts from disparate fields, explore new linkages between often isolated concepts and suggest strategies for practitioners, policymakers and researchers. Reflecting on these five lessons may help in the scaling of effective conservation actions in responsible ways to meet the triple goals of reversing biodiversity loss, combating climate change and supporting human wellbeing.

Nature would be worse off without conservation action[1], but these efforts have been inadequate to meet global biodiversity and climate targets[2]. In response, there have been widespread commitments to scaling conservation actions to bend the curve of biodiversity loss while tackling climate change[3]. Simultaneously, efforts to scale conservation actions must—at a minimum—abide by international human rights and wellbeing agreements[4]. Ultimately, the scaling of conservation actions must contribute to the triple goals of reversing biodiversity loss, responding to climate change and supporting multi-dimensional human wellbeing[5].

Yet, delivering positive outcomes at scale remains a fundamental challenge for conservation practice[6]. Scaling efforts have often fallen short, frequently failing to deliver their promised benefits and sometimes leading to adverse outcomes for people and nature. This Perspective integrates concepts and evidence from diverse fields, highlighting previously overlooked linkages between often isolated topics. It offers approaches by which practitioners, policymakers and researchers may navigate these complexities. We first discuss the mixed and sometimes problematic legacy of conservation scaling, then introduce relevant theories before offering five practical lessons

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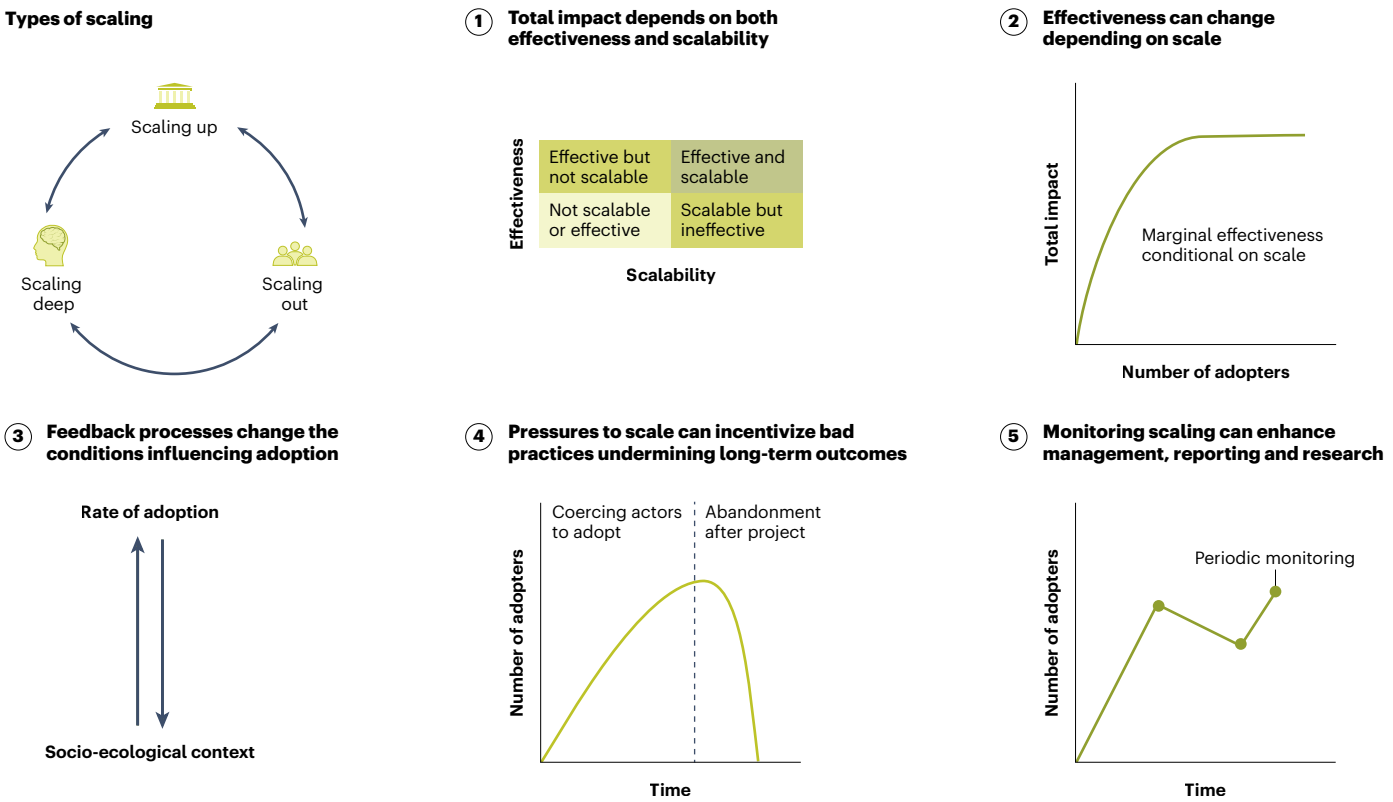


Fig. 1 | Types of scaling and lessons for scaling impact. The three types of scaling include scaling out, scaling up and scaling deep. This figure depicts the five key lessons for scaling effective conservation in responsible ways for people, nature and the climate.

at the intersection of scaling, effectiveness and social justice (Fig. 1). By encouraging those calling for scaling to reflect on these lessons, we hope to encourage a more nuanced and critical approach to going to scale for people, nature and the climate.

Mixed success of scaling for impact

On occasion, conservation actions become widely adopted with predominantly positive outcomes. For example, the adoption of community-based forest management in over 18,000 forests in Nepal appears to have decreased poverty and deforestation[7]. Similarly, hundreds of Territorial Use Rights in Fisheries areas have been established in Chile, with emerging evidence indicating that they may have delivered socio-economic benefits[8,9].

Yet, such bright spots are rare, with many widely adopted actions failing to deliver their intended benefits (for example, refs. 10–12). At least 22 Wildlife Management Areas, affecting hundreds of thousands of people, have been established in Tanzania since 2006[13,14]. Yet, Keane et al.[15] found no clear evidence that these areas have led to a widespread decrease in poverty—a primary aim. Similarly, over 350 communities have engaged in Locally Managed Marine Areas in Fiji, but evidence indicates that they have had limited effects on social and ecological outcomes (despite improvements in intermediate outcomes)[16]. Other commonly implemented actions have had substantial unintended consequences[17,18]. For example, among 111 studies of assisted migration, rewilding, biocontrol and other widely practised interventions, 36% reported unintended outcomes with sometimes persistent and severe adverse ecological effects[19].

Some projects have promoted adoption through questionable and potentially coercive approaches, undermining the long-term durability of actions. For example, Reducing Emissions from Deforestation and Forest Degradation (REDD+) initiatives have been widely implemented, but sometimes without the free, prior and informed consent of rights holders[20,21] or by misrepresenting the benefits and costs[22]. Such approaches can undermine the autonomy and livelihoods of communities (for example, ref. 23) and may also increase the risk of abandoned actions at the end of project cycles. In general, there are many instances of adoption followed by widespread abandonment. For example, community-oriented marine protected areas in Indonesia appear to have delivered short-term decreases in poverty, but these benefits were not observed after the initial donor-assisted phase ended[24]. Such outcomes are fuelled by the challenges of sustaining long-term financing following initial implementation of the project[25,26].

Scaling positive impacts is not a unique challenge to conservation but represents a cross-cutting problem in sustainable development. For instance, an evaluation of 79 large-scale irrigation projects in Sub-Saharan Africa found that only a median of 16% of their target areas were brought under irrigation[27]. According to McLean and Gargani[28], "How to enable good innovations to go to scale is one of the great perennial challenges and frontiers in development practice". The challenge of scaling sustainable development actions may partly emerge from a limited understanding of the complex and dynamic processes linking scaling, effectiveness and social justice. Reflecting on the successes and failures in past scaling efforts can provide insights into related processes, ultimately informing effective, responsible and equitable action to meet local and global conservation and sustainable development goals.

Scaling out, up and deep in theory

Conservation actions involve protecting, restoring and managing landscapes and seascapes while reducing indirect drivers of nature loss[3,29,30]. They can include direct on-the-ground activities (for example, removing invasive species) or institutional and legislative actions (for example, introducing pro-environmental policies)[31]. They can emerge from local communities, non-governmental organizations, academics and other actors and may diffuse organically without intentional

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Table 1 | Examples of conservation actions that could be scaled out, up and deep to protect, manage and restore nature

	Scaling out: expanding an action to more people or places	Scaling up: engaging with higher institutional levels to facilitate action adoption	Scaling deep: changing underlying norms, attitudes and values to inspire people to live differently
Protect	Expansion in the number of community conservation areas	National governments establishing community conservation funding mechanisms	Integrating environmental education into school curricula to cultivate a conservation stewardship ethic
Manage	Increasing the number of building development projects following biodiversity net gain (BNG) principles	Introducing government regulations that require developers to demonstrate BNG and investing in government capacity for implementation and monitoring	Working with industry associations to raise awareness of BNG and provide training to developers and local authorities
Restore	Increasing the number of farmers practising riparian restoration	Establishing policies that incentivize habitat restoration on private farms	Conservation agencies adopting rights-based approaches, supporting locally led restoration efforts

promotion or be actively encouraged (the latter being our primary focus). Furthermore, conservation actions often encompass a constellation of activities, processes and principles[32]. Thus, scaling an action might involve emulating a core set of activities or principles in new settings, but rarely involves exact replication.

Many frameworks describe the scaling of conservation and other sustainability actions[33]. Among these, Moore et al.[34] distinguish between scaling out, scaling up and scaling deep, which can be applied to conservation actions (Table 1).

Scaling out involves expanding an action to more people or places—often starting from pilot or grassroots projects—and may be achieved in several ways[33,35]: (1) by expanding an intervention across a wider programme area (for example, reproducing an action in new locations and communities)[6,35,36]; (2) by actively increasing the participatory scope of an action (for example, recruiting more households within the action area); and (3) by promoting the organic spread of actions outside a programme area (for example, through peer-to-peer learning and cross-community meetings)[6,35,36]. Scaling up involves engaging with higher institutional levels to change the rules, logics and incentives to facilitate uptake (for example, creating dedicated funding mechanisms to encourage uptake of an action)[33,34,37]. Furthermore, scaling up might involve leveraging policies and technologies within existing systems to encourage adoption at a population level[38], including through engagement with governments and the private sector[39]. Finally, scaling deep involves changing underlying attitudes, norms, knowledge and values to accelerate adoption and structural transformation[40]. These socio-psychological factors can be powerful determinants of behaviour[41] and have been the subject of substantial conservation research[42]. Although often considered relatively stable, these behavioural drivers can be influenced by well-designed interventions that draw on lessons from psychology, social marketing, sociology and other fields[43].

We use the term scalability to mean the potential for an action to scale out, up or deep, influenced by both the characteristics of an initiative and its broader socio-ecological context[44]. We describe effectiveness as a measure of an action's success or failure in achieving its stated targets, and unintended consequences as outcomes that were outside the objectives of the action. Finally, we refer to total impact as the cumulative consequences of an action at a given scale of implementation (that is, a function of its effectiveness and scale). Throughout this Perspective, we refer to systematic and evidence-based theories of scaling[45] (Box 1).

Scaling of conservation action may contribute to the wider goal of shifting social systems towards greater sustainability. Multiple frameworks describe how systems change in response to social and technological innovations[46]. For example, in the Multi-Level Perspective framework[38], innovations emerge in niches and—if successful and widely adopted—exert pressure on systems to change. If this pressure is strong enough, these systems transform, becoming the new normal. Our Perspective recognizes but does not delve deeply into the role of scaling for social transformation.

Why scale?

Some conservation actions are highly context specific; they may be effective and relevant in a given social, political and ecological setting, but are ineffective and generate unintended consequences in other contexts[47]. For example, sustainable cacao-based agroforestry might deliver desirable social and ecological outcomes in the wetter and cooler regions of the Maya Mountains of Belize but be ineffective in other places[48].

Yet, scaling can be beneficial in some circumstances. Activities, processes and principles that deliver benefits in one setting may also be advantageous in others. Thus, scaling may involve intentionally sharing, adapting and facilitating promising ideas and approaches to realize their benefits among more people and places. Potential examples include invasive vertebrate eradication techniques suitable on islands worldwide[49], delayed grassland mowing applicable across northern Europe[50] and principles for supporting rights-based approaches[51] or conservationists' wellbeing[52]. Furthermore, some actions may be more effective when implemented at larger scales (for example, ref. 53; see the section 'Effectiveness can change depending on scale'). There can also be advantages to collective and coordinated action. For instance, many actors engaging with a practice or idea—potentially mobilized by coordinating actors—can generate the critical mass needed to transform systems towards greater sustainability[54]. Therefore, the context specificity of actions and their suitability for scaling across geographical scales exists on a spectrum.

In this Perspective, we recognize the need for greater conservation action, but caution against calling for scaling in all contexts. Instead, conservation actions operating at multiple scales are needed. Moreover, we encourage those calling for scaling to interrogate why, identify appropriate scaling targets and their consequences for people and nature and assess the readiness of initiatives to go to scale (Box 2). Critically reflecting on these questions echoes calls in other sectors for responsible and deliberative innovation and scaling[55].

Five lessons for scaling the impact of conservation actions

During a series of six workshops and drawing on empirical case studies, we identified five lessons that policymakers, funders and other decision makers should consider when navigating the challenges of conservation scaling (Fig. 1). The initial four concentrate on the decision-making context, whereas the fifth is related to the availability of data essential for evidence-informed decision-making.

Total impact depends on effectiveness and scalability

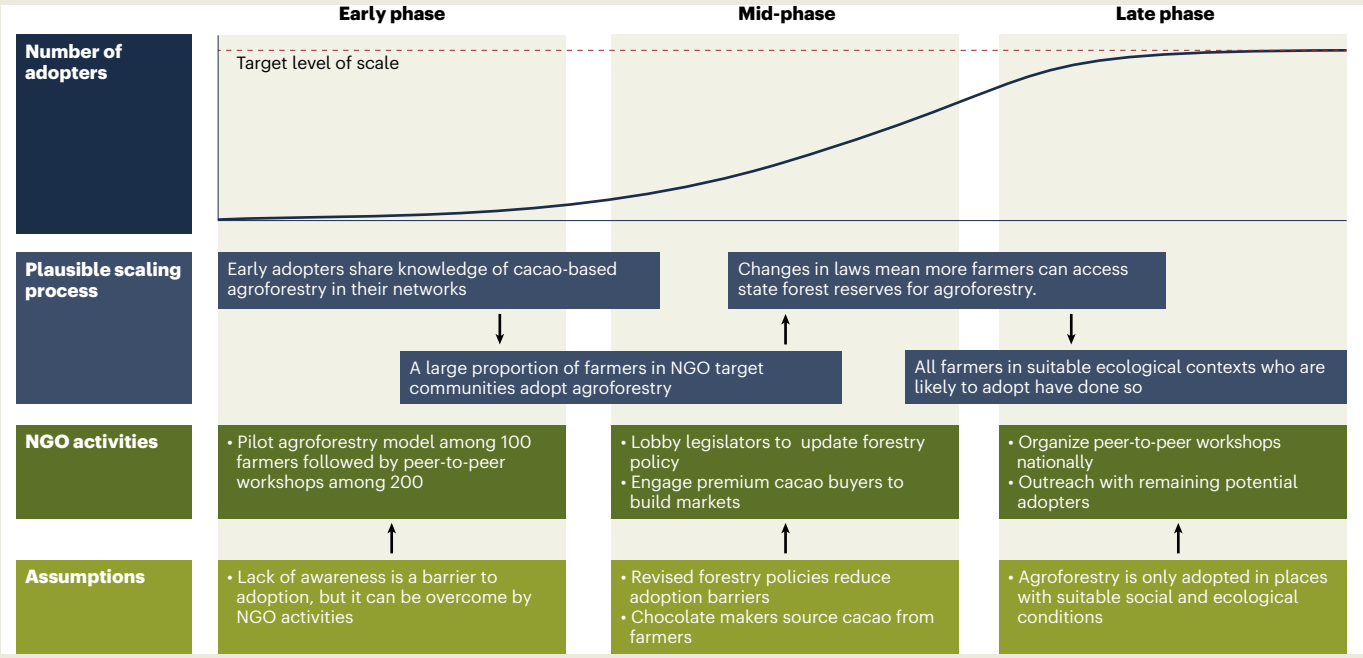
Identifying and promoting effective conservation actions will be essential for meeting global biodiversity and climate targets[56–58]. However, the total impact of an action is a function of both its effectiveness and how widely it is scaled. For instance, a hypothetical technique for replanting mangrove seedlings might be highly effective in experimental settings but require specialist knowledge and equipment, precluding

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BOX 1

Theories of scaling as tools to guide conservation scaling

Drawing on agrifood systems research, Wigboldus and Brouwers[127] argue that plans to scale agricultural innovations are often a black box of general ideas and unquestioned assumptions. In response, they propose the concept of theories of scaling[127]—a logical explanation of how a given action is expected to reach a desired scale. Theories of scaling could come in many forms depending on users' needs (see inset figure for an example) and may be standalone or integrated into broader theories of change (explaining how actions are expected to lead to envisioned changes). These theories of scaling could describe the mechanisms by which initiatives scale (for example, scaling out driven by peer-to-peer learning), facilitating activities (for example, hosting demonstration farm visits) and key assumptions (for example, learning about an initiative increases adoption). Moreover, theories of scaling could describe how these mechanisms, activities and assumptions are expected to change over a project while also being living documents that adaptively evolve. Emerging frameworks and guidance for scaling, such as those developed by Jagadish et al.[77], Sartas et al.[128] and Salafsky and Margoluis[35], may help to guide the development of these theories of scaling and their constituent activities.



A hypothetical example of a theory of scaling from the perspective of a non-governmental organization (NGO) that wishes to scale sustainable cacao-based agroforestry among farmers. The NGO chose to show the expected number of adopters across the project timeline (split into an early phase, mid-phase and late phase) to meet their scaling target. In each phase, they anticipated different mechanisms driving the scaling process, enabled by project activities with accompanying assumptions.

its widespread adoption, thus having little total impact. Recent climate mitigation research similarly differentiates between technical potential (reduction in CO_2 equivalents if an action fully realized its objectives), action feasibility (the likelihood that an actor will adopt an action) and actual mitigation (the actual CO_2 equivalents mitigated, in light of action feasibility)[59,60].

There has been substantial and much-needed progress in evaluating the effectiveness or technical potential of actions. For example, the 2021 edition of *What Works in Conservation* summarizes evidence on the effectiveness of over 2,500 conservation interventions[50]. However, less progress has been made in systematically identifying the characteristics of actions that influence their scalability (but see refs. 6,61–66) and even less has been made in integrating this understanding into conservation design, implementation and adaptive management. This lack of scaling evidence and its operationalization in conservation starkly contrasts with other sectors. For example, Piñeiro et al.[67] identified nearly 18,000 papers on the factors influencing farmers' adoption of sustainable agricultural practices. Tools such as the Adoption and Diffusion Outcome Prediction Tool (https://adopt.csiro.au) have been developed to help users to predict the rate and peak level of adoption of agricultural innovations[68]. By contrast, many conservation projects follow what Salafsky and Margoluis[35] call a pilot and pray approach (that is, supporting actions solely based on their effectiveness while ignoring their scalability risks and thus promoting approaches that deliver limited total benefit).

Conversely, supporting actions based on their scalability but overlooking evidence of their effectiveness may also undermine impact. Despite growing evidence evaluating conservation effectiveness, studies suggest that the lessons learned are often ignored when deciding which actions to promote[69–71]. For example, many large-scale tree planting programmes have failed to improve long-term ecological outcomes[10,11]. Yet, large-scale tree planting is a core approach to meeting ambitious global restoration targets (for example, the Bonn Challenge and Trillion Trees initiative). Failing to integrate effectiveness evidence into decision-making risks scaling actions based on unrealistic expectations, ultimately delivering little benefit at potentially great cost.

BOX 2

Assessing scaling readiness in conservation

Before taking steps to scale initiatives, we encourage decision makers to examine whether those initiatives are ready to scale as part of a responsible approach to conservation innovation. This reflection could involve critically reflecting on the five lessons we present below, focusing on the initiative, broader socio-ecological contexts and potential good and bad outcomes from scaling. Conservation could learn much from how other sectors have approached evaluating scaling readiness. For example, Sartas et al.[129] propose nine levels of agricultural innovation scaling readiness, ranging from an idea to an "innovation that is validated for use in an uncontrolled environment". Furthermore, they propose a scaling readiness decision support process that evaluates the innovation and innovation system, diagnoses scaling readiness, develops strategies to overcome bottlenecks, facilitates multi-stakeholder scaling processes and monitors the process for adaptive management[128]. Similar systems could be adapted to support systematic scaling processes in conservation.

Therefore, decision makers should consider and disclose evidence of both scalability and effectiveness—alongside other factors (discussed below)—when deciding which actions to pursue. One consequence of doing so is that moderately effective but scalable actions might deliver a greater overall positive impact compared with highly effective but difficult-to-scale actions. Combined consideration of scalability and effectiveness could be fostered in several ways:

(1) There is a growing but still limited systematic understanding of the factors driving the scaling of conservation actions[6]. Better theoretical understanding could help to inform the design of actions with characteristics that lend themselves towards scalability and for accessing the scaling readiness of initiatives (Box 2). Such understanding could also help funders and financers to critically evaluate the feasibility and viability of projects aiming to go to scale. Multiple theoretical frameworks have been used to understand engagement in pro-environmental behaviour[72], such as the theory of planned behaviour[73]. Most of these focus on the characteristics of individuals rather than the breadth of contextual factors that can influence adoption, thereby limiting insights into the range of strategies that could promote scaling[62]. In response, recent research has drawn on diffusion of innovation theory[44] as an integrative framework for understanding the patterns and determinants of adoption of conservation actions (for example, refs. 6,36,62,63,65,74). Pienkowski et al.[66] applied this theory to understand how the characteristics of actions, actors and socio-ecological contexts influenced spatial adoption patterns among 222,000 farmers in the Brazilian Atlantic Forest.

(2) Tools have recently been developed to help conservation practitioners to integrate effectiveness evidence into decision-making alongside other factors that influence scaling, such as feasibility and costs (for example, the Evidence-to-Decision tool[75] and economic analyses[76]). Building on guidance by Jagadish et al.[77] and Salafsky and Margoluis[35] and drawing on examples from other sectors, such as the Adoption and Diffusion Outcome Prediction Tool, analogous user-oriented tools might help practitioners to incorporate scalability considerations into project design, deployment and adaptive management.

(3) Combining effectiveness and scalability data, such tools might allow users to forecast an action's impact at different scales and at what cost, helping them to choose between options. Similar tools have been proposed to prioritize climate mitigation strategies, which account for technical potential and action feasibility[59,60]. The development of such tools is hindered by the limited availability of systematically collected and centralized data on which conservation actions have scaled and why (see the section 'Monitoring scaling can enhance evidence-informed adaptive management, reporting and research'). Moreover, it is often difficult—and sometimes impossible—to make accurate predictions within complex socio-ecological systems[78]. Clark et al.[79] applied epidemiological methods to forecast the adoption trajectory of 19 conservation case studies. These predictions were accurate under stable conditions but less accurate when there were sudden perturbations in adoption trends (for example, because of large injections of external funding). Clark et al.[79] advise against using such methods for long-term prediction, but rather to guide adaptive management towards scaling targets (see the section 'Monitoring scaling can enhance evidence-informed adaptive management, reporting and research'). Equally, methods such as agent-based modelling might help to integrate patchy empirical data with suitable theory to predict plausible patterns of future scaling. For example, Jørgensen et al.[80] used agent-based modelling to forecast business-as-usual and counterfactual scenarios of Locally Managed Marine Area adoption patterns in Fiji. These predictive methods can help to identify best bet initiatives and for participatory and scenario-based decision-making.

Effectiveness can change depending on scale

There are multiple reasons why an action's effectiveness might change depending on its scale of deployment, including issues of context specificity (see the section 'Why scale?'), unrealistic piloting, evolving support systems and feedback processes that can reinforce or hamper effectiveness.

The decision to scale conservation actions often starts from testing or observing their implementation at local scales. For example, Fiji's Locally Managed Marine Areas model incorporates and formalizes traditional communal fisheries management practices[81]. Pilot projects are often used to evaluate initiatives at small scales. However, piloting is often performed under ideal conditions, such as under the oversight of attentive project designers, with adequate financial support and in suitable socio-economic and ecological settings. For example, Massarella et al.[22] describe how pilot REDD+ projects in Tanzania received large amounts of attention and resources from civil society, funders and government agencies. However, these REDD+ projects appear to have delivered limited benefits at scale[82]. Under pilot conditions, actions may be more likely to demonstrate effectiveness, but this may decline as actions scale out to new places receiving less support and under potentially sub-optimal socio-ecological conditions. Similar issues of unrealistic piloting followed by unsuccessful scaling have been observed in the development and agricultural sectors (for example, ref. 83). Therefore, although pilot projects offer valuable learning opportunities and help to filter out undesirable actions, funders and others should recognize that they can be poor predictors of effectiveness at scale[26]. Instead, funders might ask for an evidence-informed theory of scaling (Box 1), with built-in checkpoints to evaluate pre-conditions for effectiveness at new sites (for example, suitable ecological conditions or appropriate local governance structures) or to monitor for declines in effectiveness as projects are deployed. Such theories of scaling could be developed when assessing the scaling readiness of initiatives (Box 2).

Relatedly, some initiatives may be scaled out with inadequate provision of support systems required to ensure that they deliver

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BOX 3

Experiences of scaling out, up and deep in the International Small Group and Tree Planting Program

The International Small Group and Tree Planting Program (TIST) leverages voluntary carbon markets to incentivize tree planting by smallholder farmers in Kenya, Uganda, Tanzania and India to maximize the benefits for participating farmers[130]. Since its inception in 1999, it has successfully scaled out, reaching over 170,000 participants in four countries through grassroots activity and targeted expansion, with information spreading through existing social networks[131]. This is probably due partly to strong socio-ecological feedback loops[94], whereby improved livelihoods are linked directly to the ecological health of individual farms. Participants have planted over 23 million trees and own the rights to verified carbon credits generated by quantifying their growth. This income is sufficient to offset the opportunity cost of committing to tree planting[132], but once planted, trees provide multiple co-benefits to farmers[133], including fuelwood, animal fodder, fruit or nut crops, shade and soil stabilization, providing a strong incentive for trees to be maintained over many years. The greening impact of TIST tree planting is visible at the landscape scale and has been demonstrated to extend beyond individual tree groves to neighbouring land[134].

TIST's structure facilitates rapid scaling up of innovations and best practices across the entire organization while remaining flexible and adaptive in local contexts[135]. The programme is built around small groups of six to 12 households regularly meeting to share experience and training, with a strong culture of learning by doing[131]. Information is shared rapidly between clusters via a network of facilitators and re-distributed widely in multilingual monthly newsletters[130]. Thus, in addition to the core activity of tree planting, TIST spreads a suite of complimentary agroecological practices, including tree nursery establishment, an emphasis on local seed collection and indigenous tree species, and conservation agriculture.



Meeting of TIST members in Uganda.

Rotating leadership throughout the programme structure, with equal leadership for women and men, helps to build social capital, further enhancing the programme outcomes[136], and contributes to scaling deep. Experience in leadership has empowered many participants to become more active in local communities or political processes or even to purchase land for the first time. Ensuring economic empowerment for participants of both genders[137], in turn, enables greater investment in farming, education and health[138].

benefits at scale. For example, many forest-related voluntary carbon markets have failed to deliver the claimed benefits partly because of inadequate methods to ensure additionality[12], and some biodiversity offset markets have suffered from weak governance and enforcement[84]. As such, scaling out actions may require a simultaneous scaling out, up or deep of systems and approaches to ensure their effectiveness. For example, governments might scale up policies to ensure high-quality carbon offsets accompanying the scaling out of forest-related carbon initiatives. Therefore, those developing theories of scaling should consider scaling a cluster of parallel support systems to ensure effectiveness at scale.

Economies and diseconomies of scale and collective action problems may further complicate the relationship between scale and effectiveness. For example, economies of scale mean that larger forest restoration projects may be more cost effective than smaller ones[53]. Similarly, collective action problems are common in conservation; an initiative may only become effective when a critical mass of actors engage. Projections of the net impacts of scaling actions that overlook these nonlinearities are probably wrong. Tools to forecast actions' impacts at different scales (see the section 'Total impact depends on effectiveness and scalability') should attempt to account for the most salient of these nonlinearities.

Feedback processes change socio-ecological conditions influencing future adoption

The uptake of conservation actions can fundamentally change the social and environmental conditions under which future adoption decisions are made. The adoption of conservation actions, by design, alters local environments. These environmental changes can, in turn, affect the decisions of potential adopters to engage in an action. For example, the establishment of community forestry areas by communities in Pemba Island, Zanzibar displaced fuelwood harvesting to nearby settlements (that is, leakage)[85], the inhabitants of which were in turn more likely to apply for protections on their own forests in an apparent endogenous process of scaling out[86]. Conversely, adopting conservation actions can alter environmental conditions in ways that promote abandonment, as illustrated in the evaluation of a ten-year CARE–World Wildlife Fund Alliance project in Mozambique[87]. Initially, communities supported fish and mangrove no-take zones, enhancing food security. However, as these areas regained value as sources of food and income, conservation support declined, leading to abandonment of the no-take zones[87].

Moreover, feedback reinforcement processes may exist between scaling out, up and deep, which when combined may substantially change systems[88]. For example, decentralized approaches to wildlife management were trialled at the grassroots level in southern Africa in the 1960s and 1970s[89]. These trials provided proof of concept for community-based natural resource management—a model that was consequently formalized in national policies (for example, refs. 90,91). In turn, supportive national policies and the availability of external support can facilitate further scaling[6,61]. Thus, scaling out may have reinforcement relationships with scaling up.

Similarly, scaling out might both catalyse and be driven by scaling deep. For example, the scaling deep of concerns about social outcomes

might simultaneously reinforce and been driven by the scaling out of community-based conservation approaches. Moreover, diffusion of innovation theory suggests that innovations aligned with values and beliefs are more likely to be adopted[44,77].

Meadows[92] argues that the most powerful leverage points in systems—such as shifting mindsets, goals and power dynamics—can be the most difficult to change. Nevertheless, Moore et al.[34] contend that the feedback processes between scaling out, up and deep can be harnessed to create societal transformation. For example, scaling out communal fisheries practices may lead to the scaling deep of pro-environmental values, representing deep leverage points for sustainability transitions[93]. Engaging such feedback processes may help to catalyse positive social tipping points towards sustainability[94]. Consequently, theories of scaling might include activities that combine scaling out, up and deep to transform systems (Box 3). Ultimately, Woltering et al.[26] argue that effective scaling of impact requires moving away from a narrow focus on reaching many (for example, recruiting more adopters) and towards changing the structural incentives and rules that foster engagement.

Pressures to scale can incentivize bad practices that undermine long-term outcomes

Much of the scaling research in conservation has focused on individual decision-making, such as how costs, benefits and social pressures influence adoption. However, considerable research demonstrates that conservation actions are an exercise of power[95,96]. Multiple actors at different institutional levels determine which actions are adopted and implemented on the ground, including governments, funders and financers, civil society organizations, local groups and the wider public[97–99]. In this context, money and support tend to flow from higher institutional levels, such as governments and funders, to lower levels, such as local conservation agencies and communities[100,101]. Multiple actors across these levels face pressures to scale, such as when local non-governmental organizations must commit to ambitious scaling targets to obtain funding. In this context, conservation agencies have critical roles in facilitating scaling but can be incentivized to use practices that may undermine long-term benefits for people and nature.

Emerging research highlights how supportive institutions and the availability of external support can help to facilitate the adoption of conservation actions[61]. For example, technical advice and facilitation by external actors appeared to be key drivers of landholders' adoption of Wildlife Management Units in Mexico[62]. Moreover, such external organizations can raise awareness of legal rights among local actors, facilitate access to funding or provide technical capacity.

Yet, there can be blurred boundaries between assistance and coercion, where power imbalances are manipulated to encourage local actors to adopt actions. For example, the scaling out of some community-based Wildlife Management Areas in Tanzania appears to have been driven by top-down interests, sometimes in direct opposition to those of local actors[102–104]. Facing pressures to scale, local conservation agencies may be incentivized to use a range of practices that may worsen outcomes for people and nature. For instance, conservation agencies may prioritize engaging with elites with financial and social resources that enable adoption while leveraging the status of those elites to encourage engagement by others. These elites are thus more likely to reap the early benefits of interventions, potentially widening socio-economic inequalities[77]. Conservation agencies may also misrepresent the benefits, costs and risks of adopting an action. For example, conservation organizations may have inflated residents' expectations of the benefits of REDD+ projects in Tanzania to gain buy-in, and these benefits were ultimately not delivered[22]. Moreover, local actors often experience heterogeneous benefits and costs from adopting conservation actions[85]. Thus, conservation agencies may engage in scaling discussions with local sub-groups who stand to gain while marginalizing those who might lose out. For example, conservation organizations in northern Kenya were accused of promoting community conservancies using practices that stoke intercommunity conflicts, although these allegations have been contested[105,106]. Using coercive scaling approaches risks contravening internationally recognized human rights regimens (for example, ref. 107). Moreover, many conservation actions depend on local actors' buy-in and support[108], which coercive scaling methods could weaken. Therefore, such approaches might deliver apparent short-term progress to immediate project targets (for example, enrolling more people) at the expense of providing long-term benefits.

A nuanced approach to scaling should involve scrutinizing the motives of those calling for it. Above, we describe contexts where scaling is desirable (see the section 'Why scale?'). However, calls to scale may favour some types of actors over others. Although some conservation actions scale organically, many are actively promoted by implementing organizations and funded by non-local institutions. In many contexts, these implementing organizations and higher-level institutions may have an outsized role in deciding which actions should scale[99]. Adopters—including land managers, individuals and communities—may have comparatively less say over the options presented to them and, ultimately, which actions go to scale.

Coordinating widespread implementation may require substantial financial, human and technical resources that are often only available to large organizations (often acting as brokers with smaller organizations[109]). In this context, big conservation organizations may be motivated to scale initiatives not because doing so is necessary but rather to attract commensurately large funding. Equally, funders have incentives to go big; awarding multiple small grants takes more time to assess, administer and monitor than awarding a few large ones. Moreover, private sector finance, such as via compensation-based mechanisms, is a growing revenue stream in conservation[110]. Much of this finance is directed to actions allowing private sector actors to mitigate their environmental impacts, such as nature-based carbon offsets. These actions are primarily designed to be financially competitive and meet technical requirements for measurement, calculation and monitoring, rather than to meet heterogeneous local needs[111]. Reflecting on scaling in conservation, Rocliffe and Quinlan[112] state, "For too long, market forces have compelled international conservation NGOs [...] to consolidate power rather than democratize it". In this context, unquestioned scaling narratives may marginalize small and localized conservation actors from the conservation agenda and undermine the recognition of land and resource rights. Overzealous calls for scaling may crowd-out small, locally led and effective initiatives and exclude traditional knowledge, values and practices. Ultimately, these scaling approaches may jeopardize efforts to decolonize global conservation.

A key characteristic of responsible innovation and scaling is anticipating potential consequences and impacts—both good and bad—if initiatives go to scale[55]. Therefore, well-designed theories of scaling could leverage the benefits of external assistance while integrating parallel checks, balances and systems to protect rights, such as through free, prior and informed consent. However, although the need for these systems has been recognized for decades, their use remains patchy[4]. Therefore, practitioners might co-design theories of scaling with those who will be affected, including critically reflecting on the motives and methods for promoting scaling. Moreover, funders and financers might scale out third-party verification of consent processes, establish independent grievance and outcome harvesting mechanisms that bypass implementing organizations and ensure that rights violations are penalized (for example, by suspending funding). Additionally, funders and financiers should weigh the ambition of scaling targets against the risks associated with scaling pressures and critically evaluate the feasibility of proposed theories of scaling. Ultimately, the conservation movement should join efforts to scale the decolonization of conservation. This could include scaling up Indigenous peoples' and local communities' rights recognition in national policies, scaling out approaches that foster locally led conservation (for example,

other effective area-based conservation measures[113]) and scaling deep respect for diverse knowledge, values and practices.

Monitoring scaling can enhance evidence-informed adaptive management, reporting and research

Monitoring of adoption patterns and trends is valuable for multiple reasons: it supports iterative project design and day-to-day adaptive management, can help to build an evidence base to inform scaling decisions and assists in effectively reporting progress towards global goals.

Adaptive management in conservation is the systematic and iterative process of learning from and adjusting management strategies in the face of uncertainty[114]. Conservation organizations may routinely monitor progress towards scaling milestones[115], such as the number of farmers enrolled in a programme. However, adaptive management for scaling could also involve iteratively assessing scaling readiness, identifying when further scaling becomes cost ineffective or counterproductive and adapting theories of scaling when contextual conditions change or new information comes to light. Jagadish et al.[77] draws on diffusion of innovation theory in a toolkit to help practitioners to evaluate the characteristics of actions, actors and broader contexts that influence the uptake of those actions. Such tools—potentially implemented in dialogue with prospective adopters and other stakeholders—could be used in adaptive management to stay on target towards scaling goals, as has been called for in other sectors[116].

Ideally, this adaptive management would be informed by evidence of which scaling strategies have been effective elsewhere. Most conservation organizations report data on actions to funders, and some selectively share this information publicly (for example, refs. 117,118). Yet, comprehensive, centralized and publicly available data on which conservation actions have been taken are scarce, hampering efforts to build this evidence base. Furthermore, data on adoption might not capture subsequent abandonment or intentional scaling down (for example, when they prove harmful) of actions (with the partial exception of the PADDDtracker database[119], although this lacks comprehensiveness). Abandonment might be particularly likely when funding and external support ceases, especially when this is unexpected or unplanned[120,121]. Compared with other fields, such as agriculture (for example, refs. 122–124), understanding the factors driving abandonment and their connections to adoption processes represents a major knowledge gap in conservation science.

Systematically collecting and giving researchers access to these data would strengthen understanding of scaling processes and outcomes, thus developing an evidence base to inform the scaling of decision-making. These data could be systematically collected, centralized and disclosed when governments sign agreements with local adopters (for example, the Zambian Community Forestry Management Group database[125]) or as a condition of funders issuing grants. Such systems could be integrated into emerging tools to support national reporting in the Convention on Biological Diversity, such as the Data Reporting Tool for Multilateral Environmental Agreements[126]. Broadly, we encourage conservation agencies to provide open access to data on their actions, especially when these are carried out on public land and with public funding.

Conclusion

Meeting global goals for nature, the climate and people will require substantially scaling conservation actions. Yet, efforts to scale conservation often fall short. Many scaled actions fail to meet expectations, lead to unintended harm to people and nature or are abandoned after the projects end. These examples provide potentially valuable insights, which we distil into five lessons that policymakers, funders and other decision makers could consider when scaling conservation actions. In doing so, we integrate insights and evidence from diverse fields, illustrate new connections between typically isolated concepts and propose strategies for practitioners, policymakers and researchers. First, an action's total impact depends on both its effectiveness and scalability. Substantial progress has been made in understanding the former, but effectiveness evaluations should be considered alongside robust evidence of scaling readiness (Box 2). Second, an action's effectiveness can change depending on scale because of context specificity, unrealistic piloting, evolving enabling conditions and economies and diseconomies of scale. These nonlinear dynamics are difficult to predict but could be managed through a well-designed and iteratively revised theory of scaling (Box 1). Third, feedback processes can change the socio-ecological conditions that influence future adoption. Crucially, theories of scaling that harness feedback within socio-ecological systems and processes of scaling out, up and deep could help to push actions to scale (Box 3). Fourth, pressures to scale can incentivize bad practices; coercive scaling strategies might show apparent short-term progress to project targets while harming people and undermining long-term benefits. Moreover, unquestioned scaling narratives and strategies may jeopardize progress to decolonize global conservation. Finally, monitoring scaling can enhance adaptive management, reporting and research. Understanding which actions have delivered impacts at scale requires more routine and publicly disclosed reporting of which actions have been taken, where, when and by whom (despite the challenges of doing so). Reflecting on these five lessons may help to deliver effective and equitable conservation progress to the triple goals of reversing biodiversity loss, combatting climate change and supporting multi-dimensional human wellbeing.

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Author contributions

T. Pienkowski and A. Jagadish conceived the study idea.
T. Pienkowski, A. Jagadish, W.B., G.C.B., A.P.C., A.P.E., A. Joglekar, K.S.N., T. Powell, T.W. and M.M. wrote the original draft of the manuscript. All authors reviewed and edited the manuscript and visualized the results. M.M. supervised the study. A. Jagadish and M.M. acquired the funding.

Competing interests

The authors declare no competing interests.

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